



**Generation Interconnection
System Impact Study Report
for**

Queue Project AE2-326

JACKSONS FERRY 138 KV

31.32 MW Capacity / 52.2 MW Energy

June 2020

Table Of Contents

1	Preface.....	4
2	General.....	5
2.1	Point of Interconnection	6
3	Transmission Owner Scope of Work.....	7
4	Attachment Facilities	7
5	Direct Connection Cost Estimate.....	7
6	Non-Direct Connection Cost Estimate.....	7
7	Incremental Capacity Transfer Rights (ICTRs)	8
8	Schedule.....	8
9	Interconnection Customer Requirements.....	8
10	Revenue Metering and SCADA Requirements	9
10.1	PJM Requirements	9
10.2	AEP Requirements.....	9
11	Network Impacts.....	10
12	Generation Deliverability	12
13	Multiple Facility Contingency	12
14	Contribution to Previously Identified Overloads	12
15	Potential Congestion due to Local Energy Deliverability.....	12
16	Steady-State Voltage Requirements.....	12
17	Stability and Reactive Power Requirements for Low Voltage Ride Through	13
18	Summer Peak Analysis System Reinforcements.....	13
19	Light Load Analysis	13
19.1	Generation Deliverability	13
19.2	Multiple Facility Contingency	13
19.3	Contribution to Previously Identified Overloads.....	13
19.4	Potential Congestion due to Local Energy Deliverability.....	13
19.5	Light Load Analysis System Reinforcements.....	14
19.6	Flow Gate Details.....	15
19.7	Index 1	16
20	Affected Systems	18
	LG&E.....	18

MISO.....	18
TVA.....	18
Duke Energy Progress	18
NYISO.....	18
21 Contingency Descriptions.....	19
22 Short Circuit.....	21

1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances, a generator interconnection may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection, may also contribute to the need for the same network reinforcement. Cost allocation rules for network upgrades can be found in PJM Manual 14A, Attachment B. The possibility of sharing the reinforcement costs with other projects may be identified in the feasibility study, but the actual allocation will be deferred until the impact study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

2 General

Wythe County Solar Project LLC has proposed a Storage generating facility located in Wythe County, Virginia (See Figure 2). The installed storage facilities will have a total capability of 52.2 MW with 31.32 MW of this output being recognized by PJM as Capacity. This project is an increase to the Interconnection customer's previous AC2-123 project, a 75.0 MW (44.6 MW) solar facility.

Note that AE2-326 will have separate inverters, but share the main transformer and generator lead (both with increased capacity from what was originally specified in AC2-123) and the same connection point, with the AC2-123 project. The point of interconnection will be AEP's Jacksons Ferry 138 kV substation. In addition, the Interconnection Customer has indicated that at times they will charge the batteries from the grid.

The proposed in-service date for this project is December 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AE2-326
Project Name	JACKSONS FERRY 138 KV
Interconnection Customer	Wythe County Solar Project, LLC
State	Virginia
County	Wythe
Transmission Owner	AEP
MFO	127.2
MWE	52.2
MWC	31.32
Fuel	Storage
Basecase Study Year	2022

2.1 Point of Interconnection

AE2-326 will interconnect with the AEP transmission system at Jacksons Ferry 138kV substation.

To be constructed by the Interconnection customer's previous PJM project AC2-123.

Note: It is assumed that the 138kV revenue metering system and gen lead installed for #AC2-123 will be adequate for the additional generation connection requested in AE2-326.

The AE2-326 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$0
Direct Connection Network Upgrade	\$0
Non Direct Connection Network Upgrades	\$25,000
Total Costs	\$25,000

3 Transmission Owner Scope of Work

4 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
	\$0
Total Attachment Facility Costs	\$0

5 Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
	\$0
Total Direct Connection Facility Costs	\$0

6 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Adjust the relay settings at the Jacksons Ferry 138 kV substation to accommodate the storage interconnection at the Customer Facility.	\$25,000
Total Non-Direct Connection Facility Costs	\$25,000

7 Incremental Capacity Transfer Rights (ICTRs)

None

8 Schedule

It is anticipated that the time between receipt of executed Agreements and Commercial Operation may range from 12 to 18 months if no line work is required. If line work is required, construction time would generally be between 24 to 36 months after signing Agreement execution.

9 Interconnection Customer Requirements

It is understood that Wythe County Solar Project LLC is responsible for all costs associated with this interconnection. The costs above are reimbursable to AEP. The cost of Wythe County Solar Project LLC's generating plant and the costs for the line connecting the generating plant to the Jacksons Ferry 138 kV line are not included in this report; these are assumed to be Wythe County Solar Project LLC's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for American Electric Power to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

Requirement from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 AEP Requirements

The Interconnection Customer will be required to comply with all AEP Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "Requirements for Connection of New Facilities or Changes to Existing Facilities Connected to the AEP Transmission System" document located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-aep/aep-interconnection-requirements.ashx>

11 Network Impacts

The Queue Project AE2-326 was evaluated as a 52.2 MW (Capacity 31.32 MW) injection into the Jacksons Ferry 138 kV substation in the AEP area. Project AE2-326 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-326 was studied with a commercial probability of 1.00. Potential network impacts were as follows:

Summer Peak Load Flow

12 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

13 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

14 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

15 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/D C	MW IMPACT
7445603	934750	AD1-102 TAP	138.0	AEP	242609	05CLYTR2	138.0	AEP	1	242512 05CLOVR D 765 242514 05J.FERR 765 1	operation	243.0	141.81	143.29	AC	4.16
7445604	934750	AD1-102 TAP	138.0	AEP	242609	05CLYTR2	138.0	AEP	1	Base Case	operation	219.0	121.63	123.0	AC	3.44

16 Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

To be evaluated during the Facilities Study Phase

17 Stability and Reactive Power Requirements for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be evaluated during the Facilities Study Phase

18 Summer Peak Analysis System Reinforcements

None

19 Light Load Analysis

Light Load Studies (applicable to wind, coal, nuclear, and pumped storage projects).

19.1 Generation Deliverability

(Single or N-1 contingencies for the full energy output of the interconnection)

None

19.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

19.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJE CT LOADIN G %	POST PROJE CT LOADIN G %	AC/D C	MW IMPAC T
14914535	934750	AD1-102 TAP	138.0	LL	242609	05CLYTR2	138.0	AEP	1	AEP_P4_#8478_05CLOVRD 765_BB	breaker	243.0	115.08	116.46	AC	3.93

19.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

19.5 Light Load Analysis System Reinforcements

To resolve the AD1-102 Tap – Claytor 138 kV line overload:

The existing conductor is ACSR~795~45/7~TERN. This line is sag limited so a sag study may be necessary during the Facilities Study. If the sag study determines a complete reconductor/rebuild is required, the cost estimate to reconductor/rebuild the 4.69 mile section of line is \$7,035,000. An approximate construction time would be 24 to 36 months after signing an interconnection agreement.

Note 1: Per the Z2-042 Facilities Study: a sag study has been previously completed on the Wurno – Claytor 138 kV line which includes this line section. The results of the sag study indicated the need to replace 17 structures on the Wurno – Claytor 138 kV line. Cost estimate is \$2.739M. PJM Network Upgrade N4505.1. Fewer structures may need to be replaced for just this line section. This information will need to be confirmed during the Facilities Study.

Note 2: This overload is presently driven by a prior queue cycle (AD1-102) and therefore AE2-326 presently does not receive a cost allocation to this upgrade assuming the upgrade cost is less than \$5M per PJM cost allocation rules. As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, or the upgrade scope and cost is determined to cost more than \$5M, Queue Project AE2-326 could receive cost allocation.

Note 3: Although Queue Project AE2-326 may not have cost responsibility for this upgrade, Queue Project AE2-326 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-326 comes into service prior to completion of the upgrade, Queue Project AE2-326 will need an interim study.

19.6 Flow Gate Details

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

19.7 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
14914535	934750	AD1-102 TAP	LL	242609	05CLYTR2	AEP	1	AEP_P4_#8478_05CLOVRD 765_BB	breaker	243.0	115.08	116.46	AC	3.93

Bus #	Bus	MW Impact
913031	Y1-006 C	0.3215
913032	Y1-006 E	2.1515
934751	AD1-102 C	12.5063
934752	AD1-102 E	83.7012
937492	AD2-205 E	0.2332
943031	AE2-326 C	2.0071
943032	AE2-326 E	1.3349

Affected Systems

20 Affected Systems

LG&E

None

MISO

None

TVA

None

Duke Energy Progress

None

NYISO

None

21 Contingency Descriptions

Contingency Name	Contingency Definition
Base Case	
242512 05CLOVRD 765 242514 05J.FERR 765 1	CONTINGENCY '242512 05CLOVRD 765 242514 05J.FERR 765 1' / 8408 OPEN BRANCH FROM BUS 242512 TO BUS 242514 CKT 1 / 242512 05CLOVRD 765 242514 05J.FERR 765 1 END
AEP_P4_#8478_05CLOVRD 765_BB	CONTINGENCY 'AEP_P4_#8478_05CLOVRD 765_BB' OPEN BRANCH FROM BUS 242512 TO BUS 242519 CKT 14 / 242512 05CLOVRD 765 242519 05CLOVRD 500 14 OPEN BRANCH FROM BUS 242512 TO BUS 242524 CKT 10 / 242512 05CLOVRD 765 242524 05CLOVRD 345 10 END

Short Circuit

22 Short Circuit

The following Breakers are overduty:

Jacksons Ferry 138 kV P2

PJM Supplemental Project s1382 will replace breaker P2 at Jacksons Ferry 138kV.

Figure 1: AE2-326 Jacksons Ferry 138kV Substation
One Line Diagram

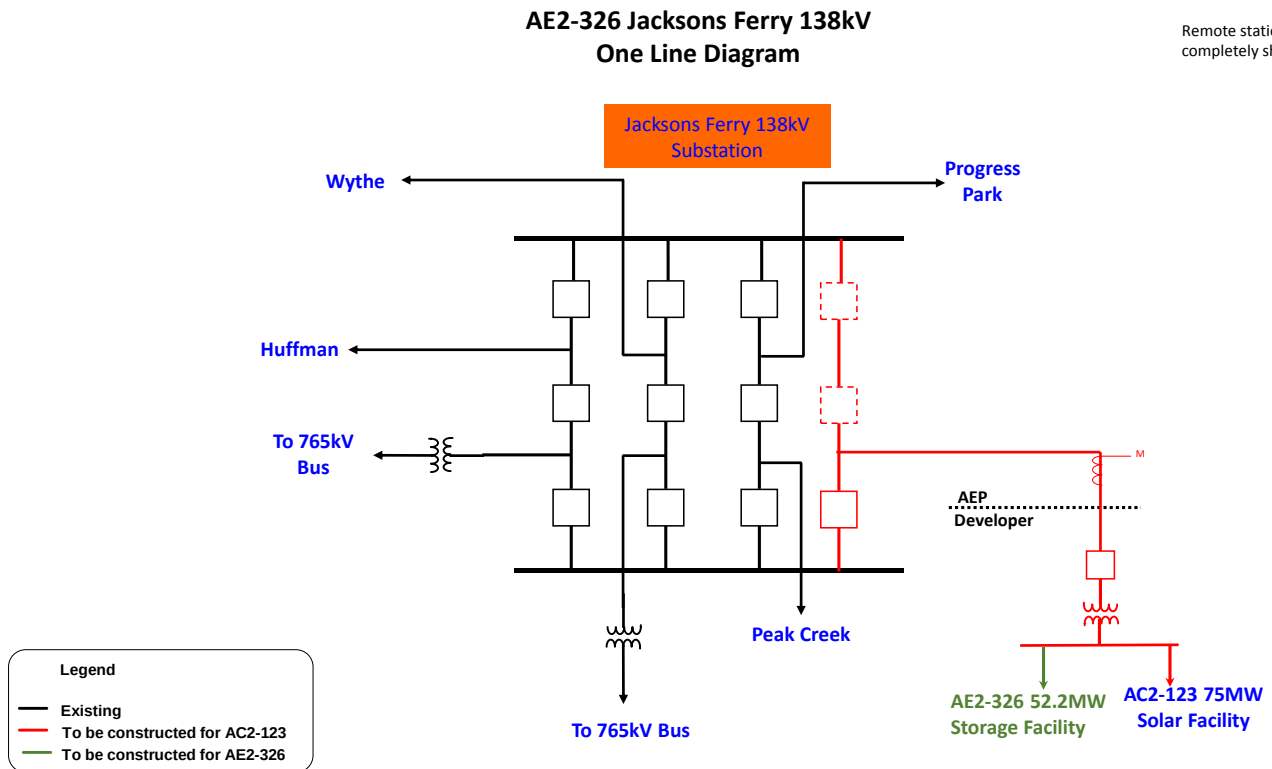


Figure 2: Point Of Interconnection (Jacksons Ferry 138kV)

